

Please write clearly in block capitals.

Centre number

--	--	--	--	--

Candidate number

--	--	--	--

Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL AS CHEMISTRY (9620)

Unit 2: Organic 1 and Physical 1

Tuesday 9 January 2024 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 70.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	



J A N 2 4 C H O 2 0 1

Answer **all** questions in the spaces provided.

0 1

Alkanes are used as fuels in internal combustion engines in cars.

0 1 . 1

Write an equation to show the incomplete combustion of heptane (C_7H_{16}) to form carbon monoxide and water only.

[1 mark]

0 1 . 2

Some of alkane **A** is found unburned in car exhaust fumes. Analysis shows that 0.0381 mol of **A** has a mass of 4.35 g

Determine the molecular formula of **A**.
Show your working.

[2 marks]

Molecular formula _____

0 1 . 3

Give **one** environmental reason why unburned alkanes are pollutant gases.

[1 mark]

0 1 . 4

Combustion of alkane fuels can produce sulfur dioxide.

Give **one** environmental reason for removing sulfur dioxide from the gaseous products of fuel combustion.

[1 mark]



Nitrogen monoxide and carbon monoxide are also produced during the combustion of alkane fuels. These pollutant gases can be removed using a catalytic converter.

Do not write
outside the
box

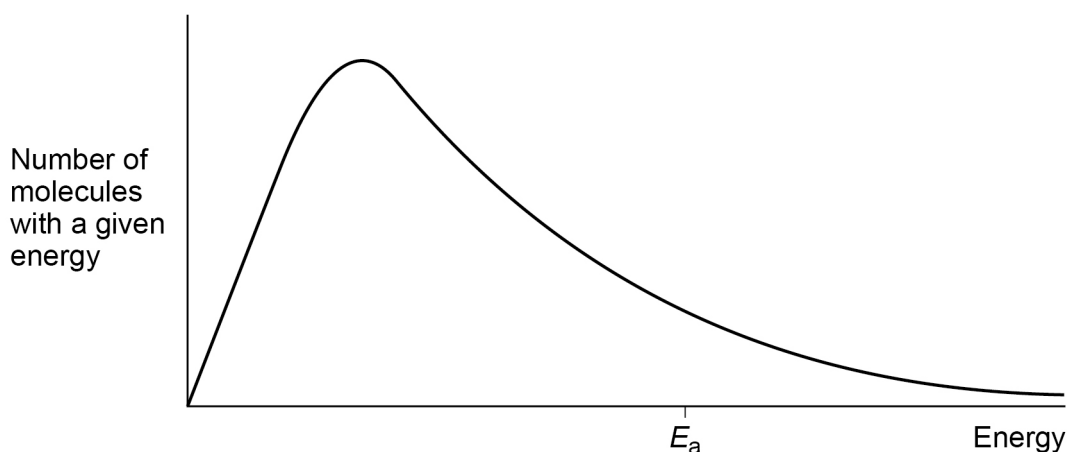
0 1 . 5

Suggest why the reaction is slow when a catalyst is **not** used.

[1 mark]

Figure 1 shows a Maxwell–Boltzmann distribution for a gaseous mixture of nitrogen monoxide and carbon monoxide at a given temperature.

Figure 1



0 1 . 6

State the effect, if any, to the **area** under the curve when a catalyst is added to this gaseous mixture.

[1 mark]

0 1 . 7

Explain how a catalyst increases the rate of a reaction.

[3 marks]

0 1 . 8

Identify a catalyst used in the catalytic converter.

[1 mark]



0 2

The hydrocarbons in crude oil are separated into fractions by fractional distillation.

0 2 . 1

State the meaning of fractions.

[1 mark]

There are three isomers with the molecular formula C_5H_{12}

0 2 . 2

Explain why dimethylpropane has the lowest boiling point.

[2 marks]

0 2 . 3

Draw skeletal formulas of the **two** other isomers with molecular formula C_5H_{12}

[2 marks]

Isomer 1

Isomer 2

0 2 . 4

Some of the hydrocarbons separated by fractional distillation are cracked.
One molecule of a hydrocarbon **B** is cracked to produce two molecules of ethene,
two molecules of propene and one molecule of butane.

Give the molecular formula of hydrocarbon **B**.

State the type of cracking involved.

Suggest why the percentage atom economy for this reaction is 100%

[3 marks]

Molecular formula of **B** _____

Type of cracking _____

Atom economy reason _____



0	3
---	---

This question is about addition polymers.

0	3	.	1
---	---	---	---

The polymer poly(propene) is produced from propene.

Draw the repeating unit of poly(propene).

[1 mark]

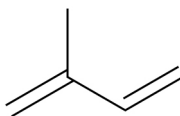
0	3	.	2
---	---	---	---

Give **one** reason why poly(propene) is unreactive.

[1 mark]

0	3	.	3
---	---	---	---

An addition polymer is formed from isoprene.
The skeletal formula of isoprene is shown.



Use IUPAC rules to name isoprene.

[1 mark]

3

Turn over ►

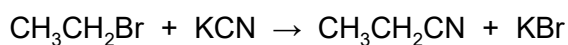


0	4
---	---

This question is about halogenoalkanes.

0	4	.	1
---	---	---	---

Bromoethane reacts with a warm aqueous ethanolic solution of potassium cyanide in a nucleophilic substitution reaction.



Outline a mechanism for the reaction.

[2 marks]

0	4	.	2
---	---	---	---

State why the reaction of potassium cyanide with iodoethane is faster than the reaction of potassium cyanide with bromoethane.

[1 mark]

0	4	.	3
---	---	---	---

2-Bromopropane reacts with a hot ethanolic solution of potassium hydroxide to form propene.

State the role of the hydroxide ion in this reaction.

Write an overall equation for the reaction.

[2 marks]

Role of the hydroxide ion _____

Overall equation

5

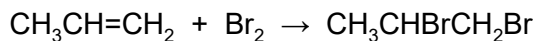


0 5

This question is about the formation of halogenoalkanes.

0 5 . 1

Propene reacts with bromine.



Name and outline the mechanism of this reaction.

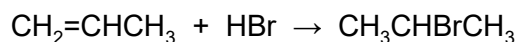
[4 marks]

Name of mechanism _____

Mechanism _____

0 5 . 2

Propene reacts with HBr to form two possible products. The equation shows the formation of the major product.



State the type of intermediate in the formation of the major product.

Explain why this intermediate is more stable than the intermediate that forms the minor product.

[3 marks]

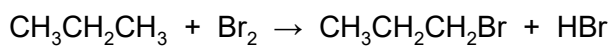
Type of intermediate _____

Explanation _____

Turn over ►

0	5	.	3
---	---	---	---

Propane can react with bromine in ultraviolet light to form $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$



State the role of ultraviolet light in this reaction.

Outline **two** propagation steps from the mechanism.

[3 marks]

Role of ultraviolet light _____

Propagation step 1 _____

Propagation step 2 _____

10



Turn over for the next question

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►



0	6
---	---

This question is about alcohols.

0	6	.	1
---	---	---	---

Alcohol **S** contains 60.0% C, 13.3% H and 26.7% O

Determine the empirical formula of **S**.

[2 marks]

Empirical formula _____

0	6	.	2
---	---	---	---

Alcohol **S** has $M_r = 60.0$

Alcohol **S** can be oxidised to form a compound that gives a signal at 1715 cm^{-1} in its infrared spectrum. The compound does **not** react with Tollens' reagent.

Draw the displayed formula of alcohol **S**.

[1 mark]

0	6	.	3
---	---	---	---

A position isomer of alcohol **S** is oxidised to give compound **T**.
T reacts with NaHCO_3

Identify **T**.

State what is observed when **T** reacts with NaHCO_3

[2 marks]

Identity of **T** _____

Observation _____



0 6 . 4

Identify a test-tube reagent, or combination of reagents, that can be used to oxidise primary and secondary alcohols but not tertiary alcohols.

State the colour change that is observed.

[2 marks]

Reagent(s) _____

Colour change _____

0 6 . 5

A different alcohol $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{OH}$ is oxidised to form compound **X**.
X reacts with Fehling's solution.

Write an equation for this oxidation of $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{OH}$
Use [O] to represent the oxidising agent.

Use IUPAC rules to name **X**.

State what is observed when **X** reacts with Fehling's solution.

[3 marks]

Equation

Name of **X** _____

Observation _____

0 6 . 6

Ethanol reacts with epoxyethane to form compound **Z** ($\text{C}_4\text{H}_{10}\text{O}_2$)

State the C–O–C bond angle in epoxyethane.

Give the structure of **Z**.

[2 marks]

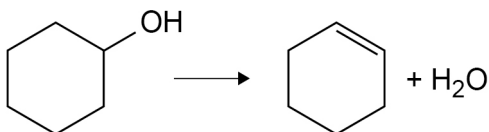
C–O–C bond angle in epoxyethane _____

Structure of **Z**



0 7

Cyclohexene can be made by adding an excess of concentrated sulfuric acid to hot cyclohexanol.



0 7 . 1

Name and outline the mechanism for this reaction.

[4 marks]

Name of mechanism _____

Mechanism

0 7 . 2

Identify a reagent, or combination of reagents, that can be used to confirm that an alkene is formed.

State what is observed.

[2 marks]

Reagent _____

Observation _____

Table 1 contains data about cyclohexene and cyclohexanol.

Use the information in **Table 1** to help answer Questions **07.3–07.5**.

Table 1

Compound	Boiling point / °C	Density / g cm ⁻³	<i>M_r</i>
Cyclohexene	82.8	0.811	82.0
Cyclohexanol	161.8	0.948	100.0



07.3

Name a technique that can be used to separate cyclohexene from unreacted cyclohexanol.

[1 mark]

07.4

Explain, using the information in **Table 1**, how the purity of the separated cyclohexene can be determined.

[1 mark]

07.5

In an experiment 15.0 cm^3 of cyclohexanol forms 9.62 g of cyclohexene.

Calculate the maximum mass of cyclohexene that can be made from 15.0 cm^3 of cyclohexanol.

Calculate the percentage yield of cyclohexene in this experiment.

Use data from **Table 1** to answer this question.

[4 marks]

Maximum mass of cyclohexene _____ g

Percentage yield of cyclohexene _____

12

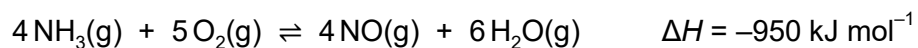
Turn over ►



0 8

This question is about chemical equilibria.

The equation for the reaction between ammonia and oxygen is

**0 8 . 1**

Write an expression for K_c for this reaction.

Give the units of K_c

[2 marks] K_c

Units _____

0 8 . 2

In an experiment 0.650 mol of NH_3 and 0.743 mol of O_2 are mixed and left to reach equilibrium.

The equilibrium mixture contains 0.144 mol of NO and 0.216 mol of H_2O

Calculate the amount, in mol, of NH_3 and O_2 in the equilibrium mixture.

[2 marks]Mol NH_3 _____Mol O_2 _____

0 8 . 3

Explain why the K_c value for this reaction decreases as the temperature increases.

[3 marks]

0 8 . 4

Explain why using a catalyst does **not** affect the equilibrium yield of NO

[1 mark]

0 8 . 5

The table shows three different gaseous equilibria reactions all at the same temperature.

Tick (✓) **one** box to select the reaction where an increase in overall pressure increases the equilibrium yield of **C**.

[1 mark]

Reaction
$A + 2B \rightleftharpoons 4C$
$A + B \rightleftharpoons 2C$
$2A + B \rightleftharpoons C$

☐
☐
☐

9

END OF QUESTIONS



There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**



There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Copyright information

For confidentiality purposes, all acknowledgements of third-party copyright material are published in a separate booklet. This booklet is published after each live examination series and is available for free download from www.oxfordaqa.com

Permission to reproduce all copyright material has been applied for. In some cases, efforts to contact copyright-holders may have been unsuccessful and OxfordAQA will be happy to rectify any omissions of acknowledgements. If you have any queries please contact the Copyright Team.

Copyright © 2024 OxfordAQA International Examinations and its licensors. All rights reserved.



2 0



2 4 1 X C H 0 2

IB/G/Jan24/CH02